

3.3 Organic chemistry

3.3.1 Introduction to organic chemistry

Organic chemistry is the study of the millions of covalent compounds of the element carbon.

These structurally diverse compounds vary from naturally occurring petroleum fuels to DNA and the molecules in living systems. Organic compounds also demonstrate human ingenuity in the vast range of synthetic materials created by chemists. Many of these compounds are used as drugs, medicines and plastics.

Organic compounds are named using the International Union of Pure and Applied Chemistry (IUPAC) system and the structure or formula of molecules can be represented in various different ways. Organic mechanisms are studied, which enable reactions to be explained.

In the search for sustainable chemistry, for safer agrochemicals and for new materials to match the desire for new technology, Chemistry plays the dominant role.

3.3.1.1 Nomenclature

Content	Opportunities for skills development
Organic compounds can be represented by: • empirical formula • molecular formula • general formula • structural formula • displayed formula • skeletal formula. The characteristics of a homologous series, a series of compounds containing the same functional group. IUPAC rules for nomenclature. Students should be able to: • draw structural, displayed and skeletal formulas for given organic compounds • apply IUPAC rules for nomenclature to name organic compounds limited to chains and rings with up to six carbon atoms each • apply IUPAC rules for nomenclature to draw the structure of an organic compound from the IUPAC name limited to chains and rings with up to six carbon atoms each.	

3.3.1.2 Reaction mechanisms

Content	Opportunities for skills development
Reactions of organic compounds can be explained using mechanisms. Free-radical mechanisms: - the unpaired electron in a radical is represented by a dot - the use of curly arrows is not required for radical mechanisms. Students should be able to: - write balanced equations for the steps in a free-radical mechanism. Other mechanisms: - the formation of a covalent bond is shown by a curly arrow that starts from a lone electron pair or from another covalent bond - the breaking of a covalent bond is shown by a curly arrow starting from the bond. Students should be able to: - outline mechanisms by drawing the structures of the species involved and curly arrows to represent the movement of electron pairs.	

3.3.1.3 Isomerism

Content	Opportunities for skills development
Structural isomerism. Stereoisomerism. <i>E–Z</i> isomerism is a form of stereoisomerism and occurs as a result of restricted rotation about the planar carbon–carbon double bond. Cahn–Ingold–Prelog (CIP) priority rules. Students should be able to: - define the term structural isomer - draw the structures of chain, position and functional group isomers - define the term stereoisomer - draw the structural formulas of <i>E</i> and <i>Z</i> isomers - apply the CIP priority rules to <i>E</i> and <i>Z</i> isomers.	MS 4.2 Students could be given the structure of one isomer and asked to draw further isomers. Various representations could be used to give the opportunity to identify those that are isomeric. MS 4.1, 4.2 and 4.3 Students understand the origin of <i>E–Z</i> isomerism. Students draw different forms of isomers.